

SYNTHETIC COGNITION

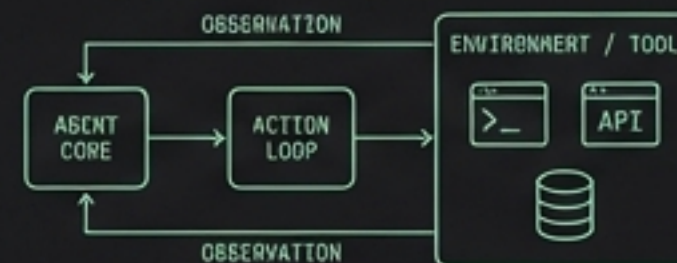
From Transformer Architecture to Autonomous Agents

1. The Engine: Tokenization & Transformers



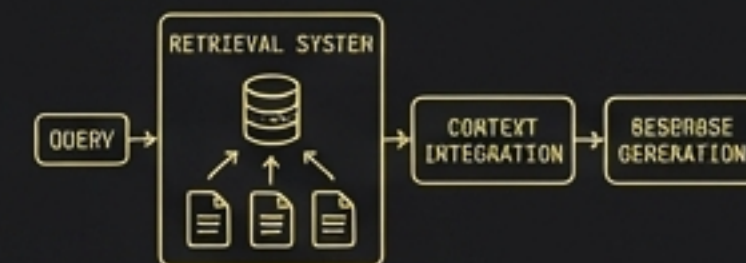
The core processing unit, converting input into structured data tokens and utilizing attention mechanisms for understanding and generation.

2. The Agency: The OpenAI SDK & Action Loops



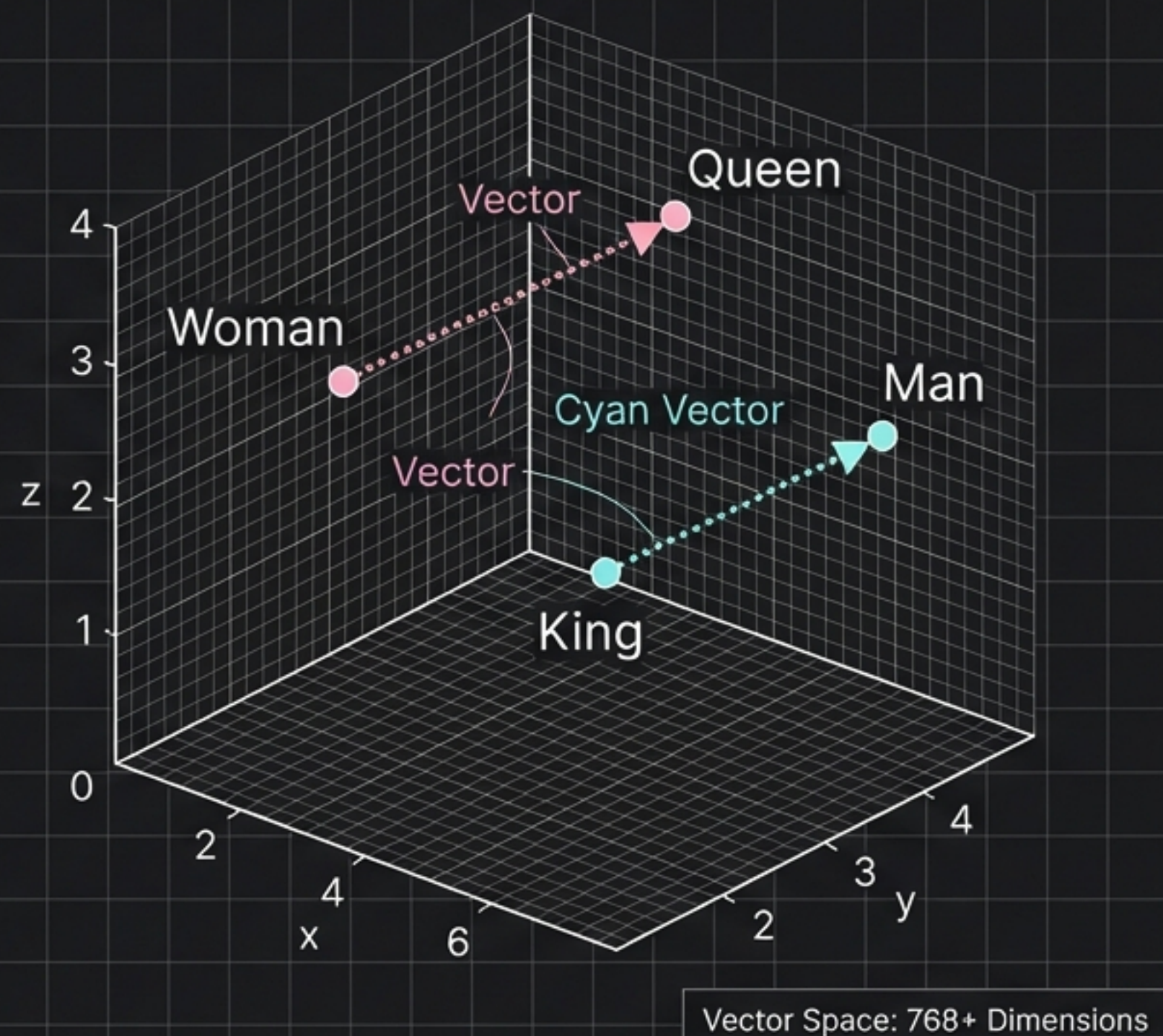
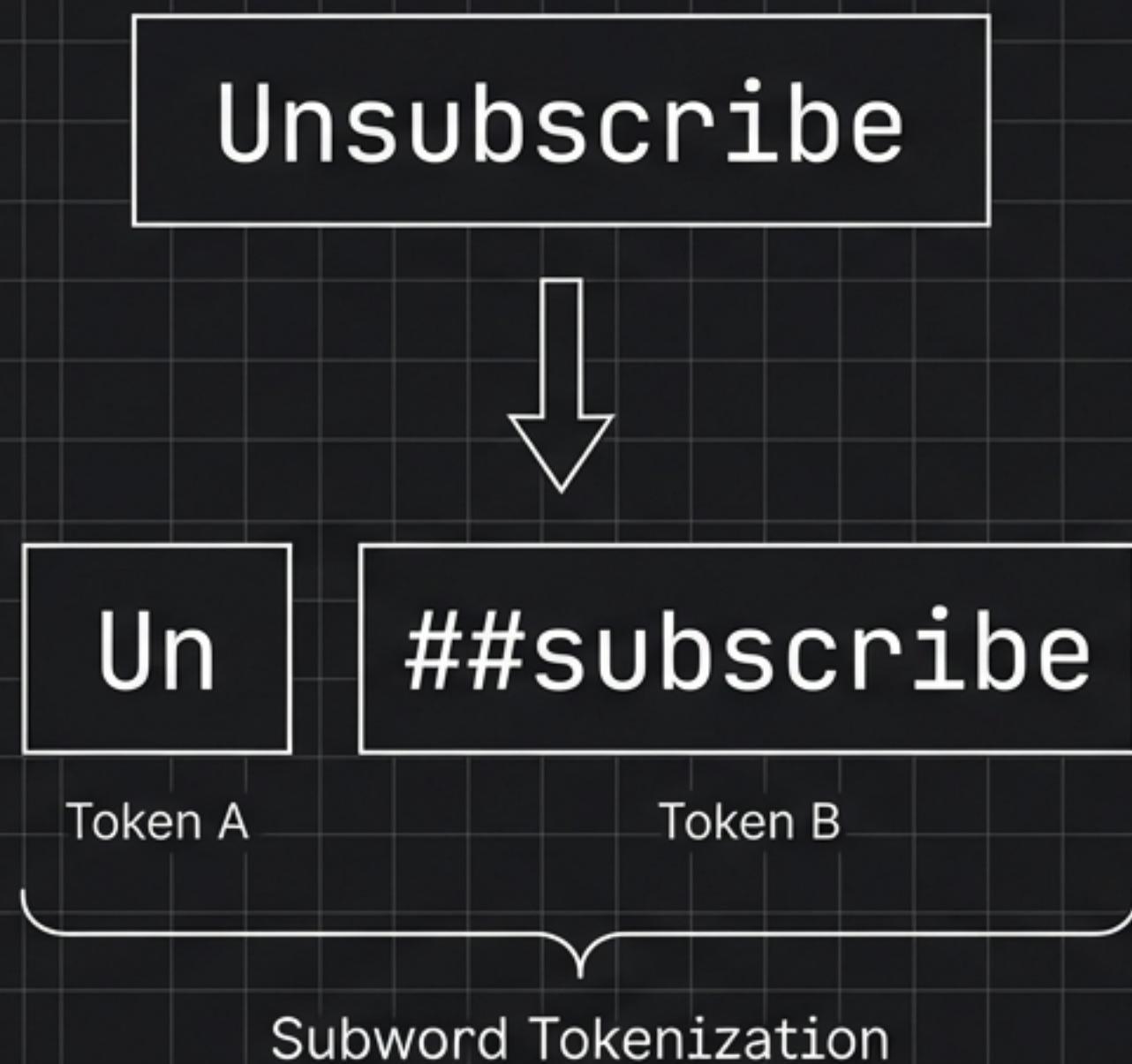
Enabling autonomous interaction with the world through tool use, decision-making, and iterative feedback loops.

3. The Context: Advanced Agentic RAG

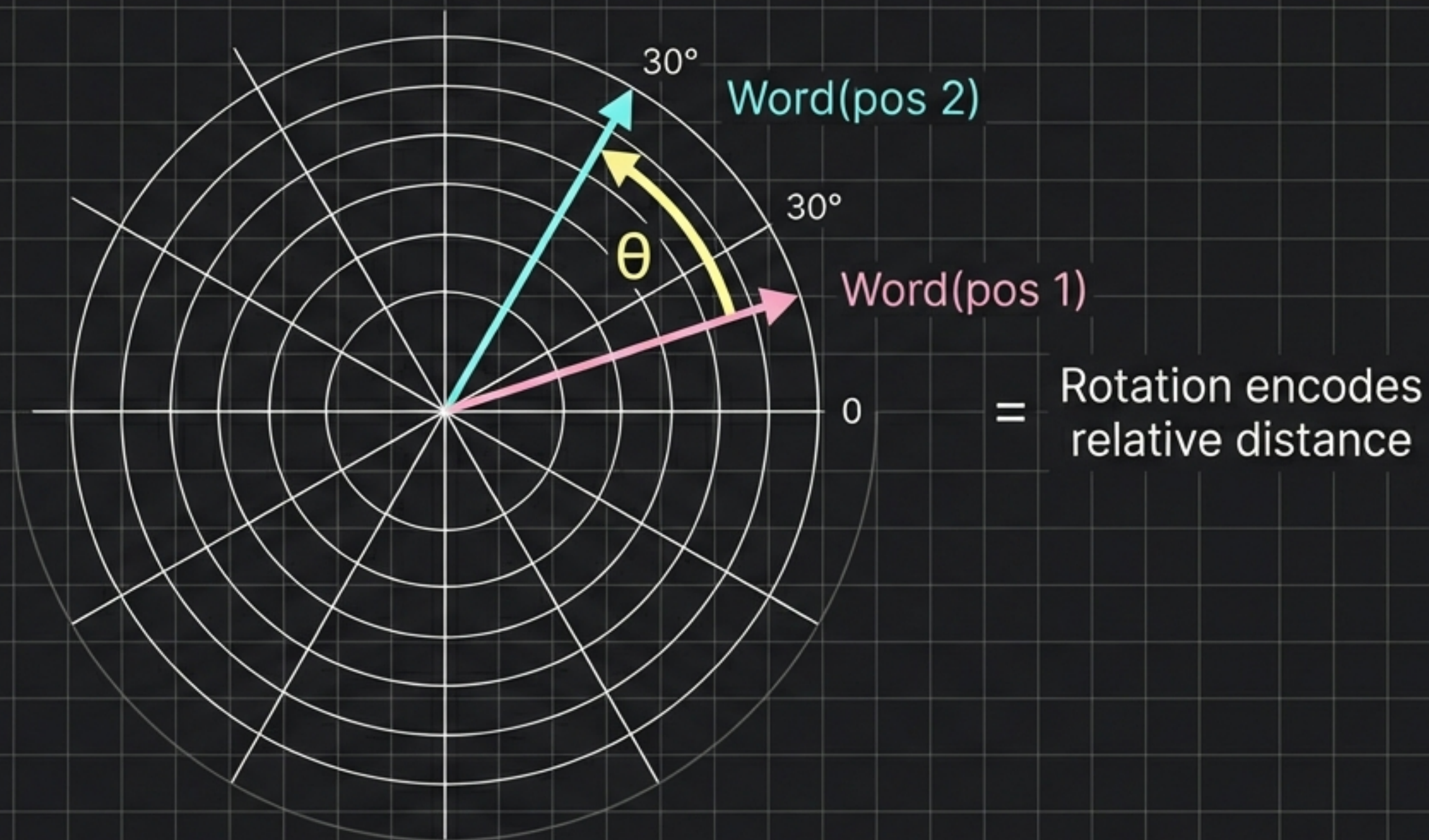


Integrating external knowledge and long-term memory into the agent's decision-making process for enhanced accuracy.

THE ATOMIC UNIT: TOKENS & EMBEDDINGS



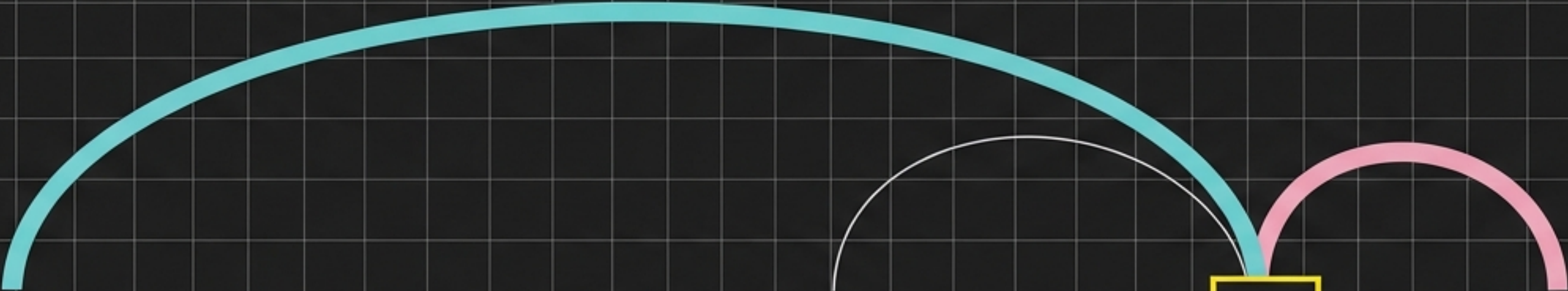
PRESERVING ORDER: RoPE



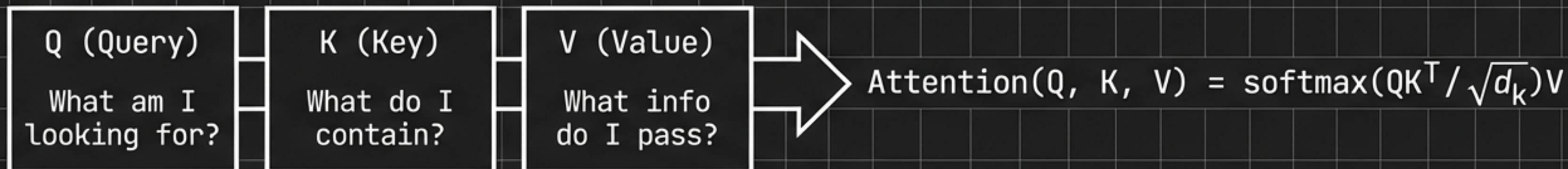
Rotary Positional Embeddings allow the model to understand sequence without fixed absolute positions.

THE MECHANISM OF MEANING: SELF-ATTENTION

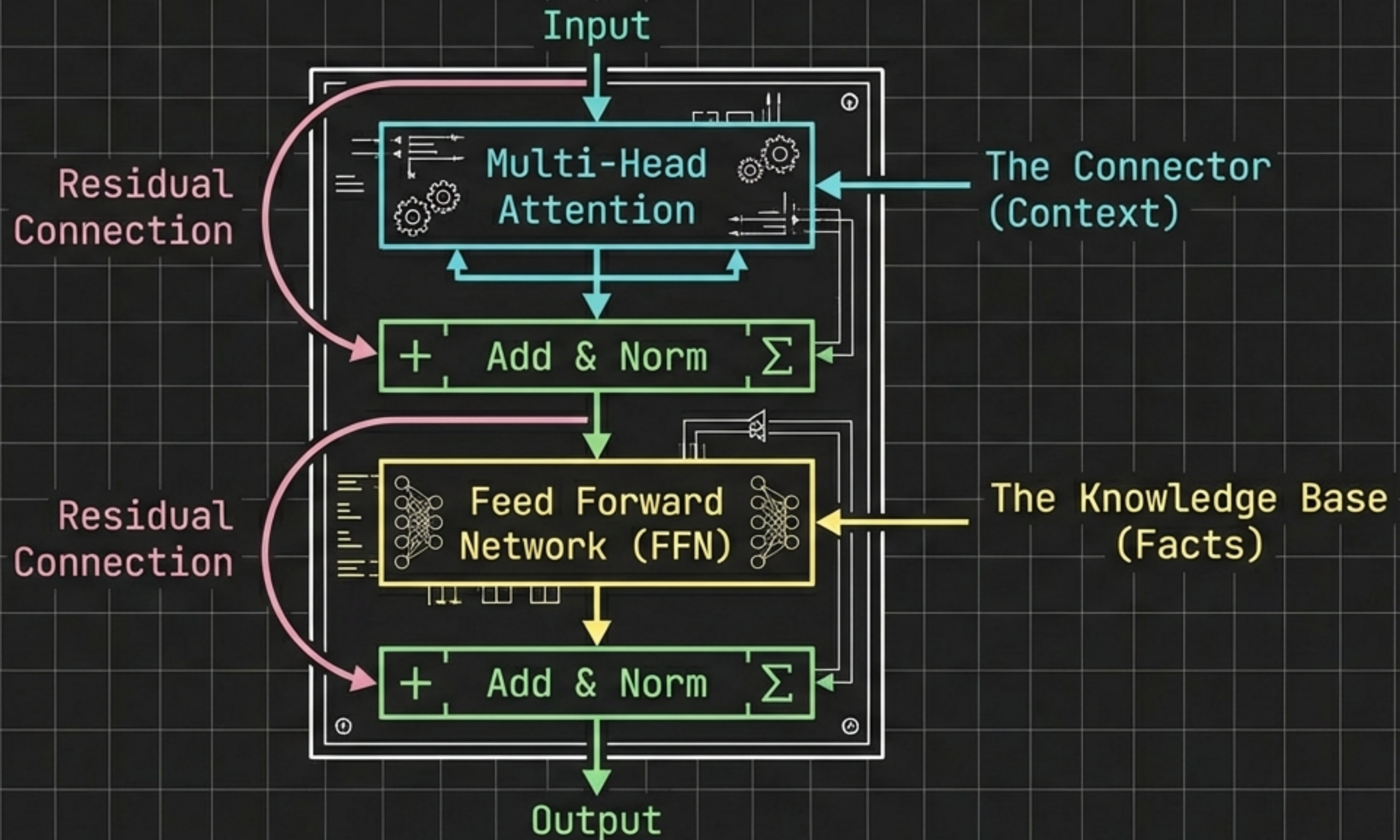
The animal didn't cross the street because **it** was tired.



"Dinner Party" matrix diagram

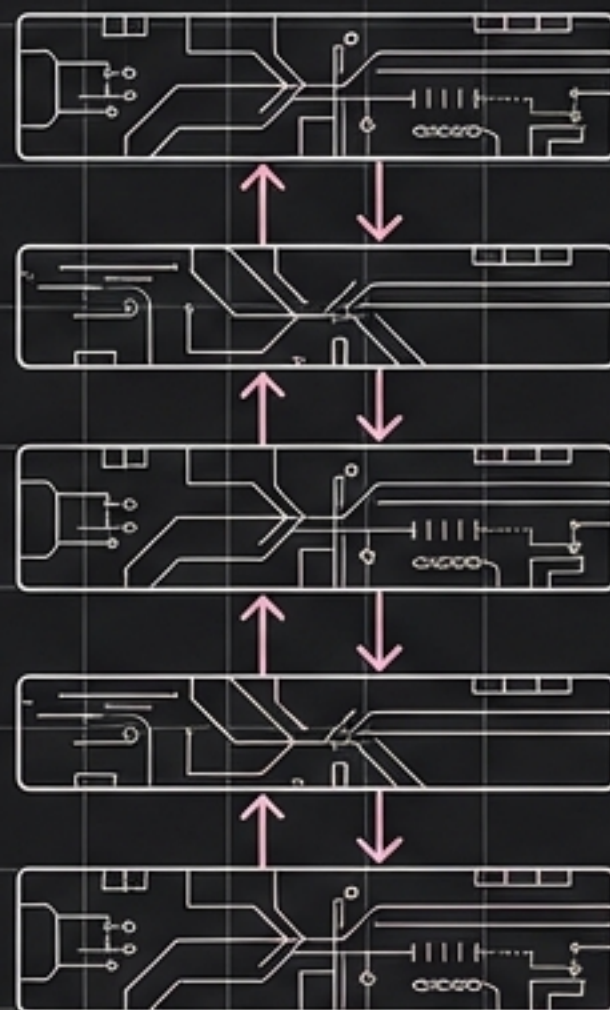


THE TRANSFORMER BLOCK



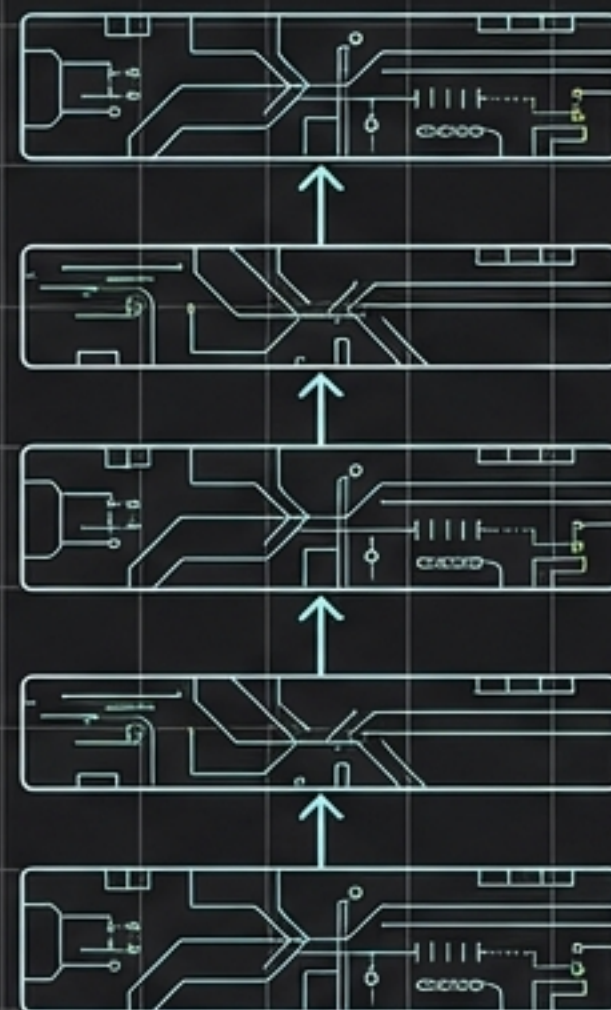
ARCHITECTURE VARIANTS

ENCODER (BERT)



Bidirectional context.
Good for classification.

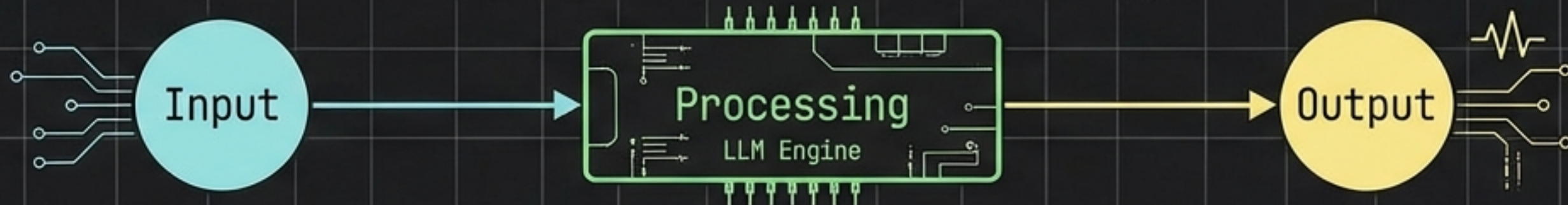
DECODER (GPT)



Unidirectional / Causal.
Good for generation & agents.

THE AGENTIC SHIFT

PREDICTION (Standard LLM)



AGENCY (The ReAct Loop)



CODE: THE FOUNDATION

```
CODE

from OpenAI import OpenAI

client = OpenAI()
completion = client.chat.completions.create(
    model="gpt-4o",
    # The Persona Setup
    messages = [
        {"role": "system",
         "content": "You are a helpful assistant..."},
        {"role": "user",
         "content": "Calculate  $3x + 11 = 14$ "}
    ]
)
```

Defining the
Agent Persona

CODE: DEFINING TOOLS

```
CODE

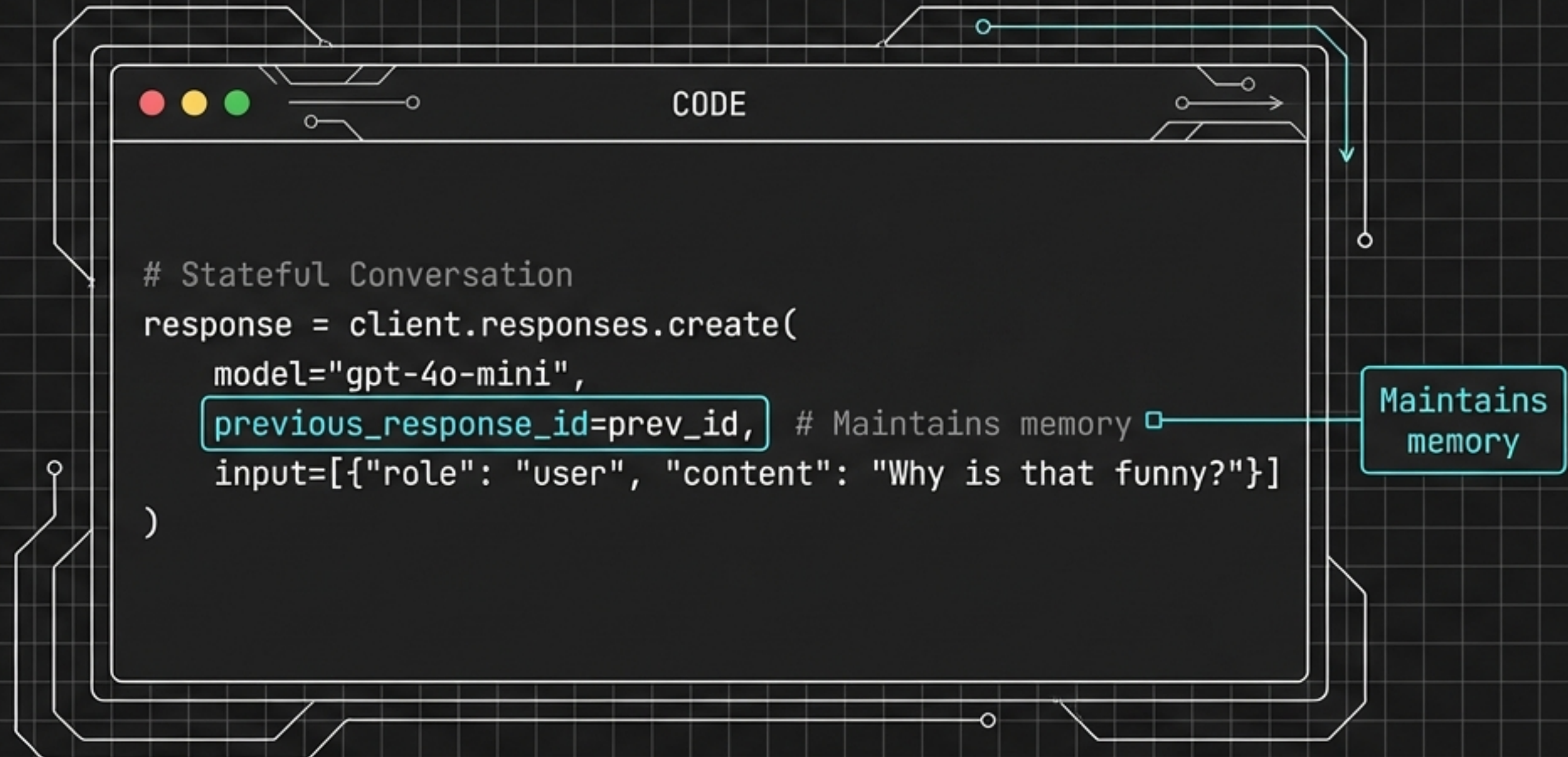
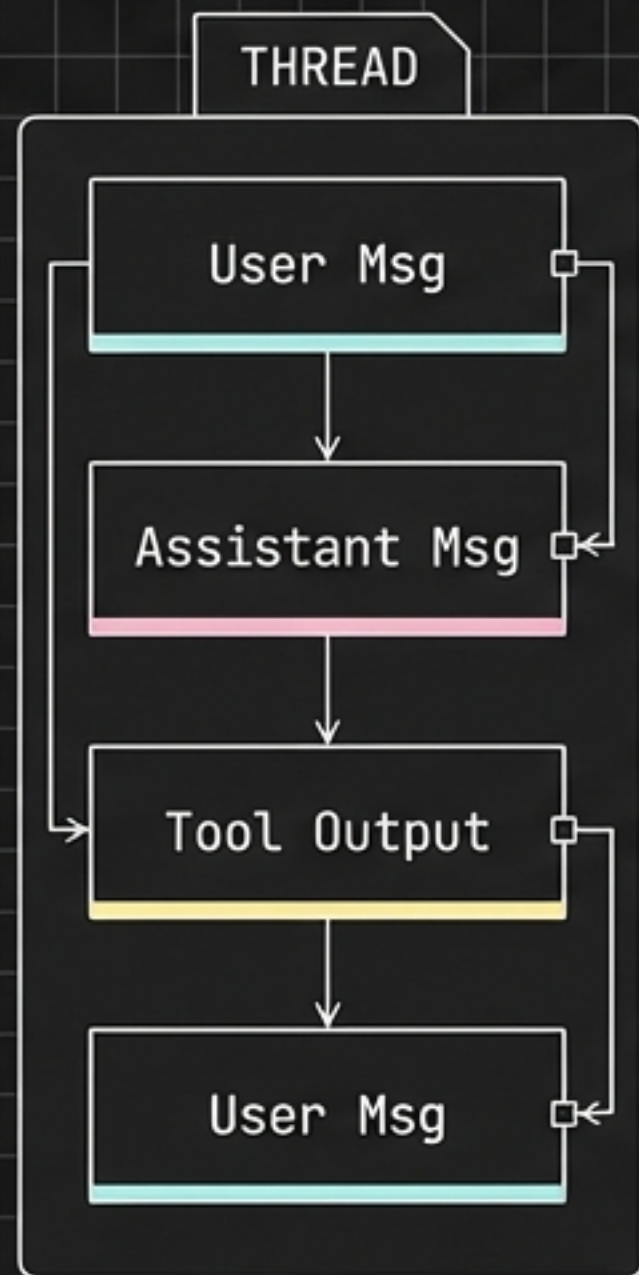
@tool
def get_weather(city: str) -> str:
    """Returns current weather for a specific city."""
    return f"The weather in {city} is 24°C."

@tool
def add_numbers(a: int, b: int) -> int:
    """Adds two numbers."""
    return a + b

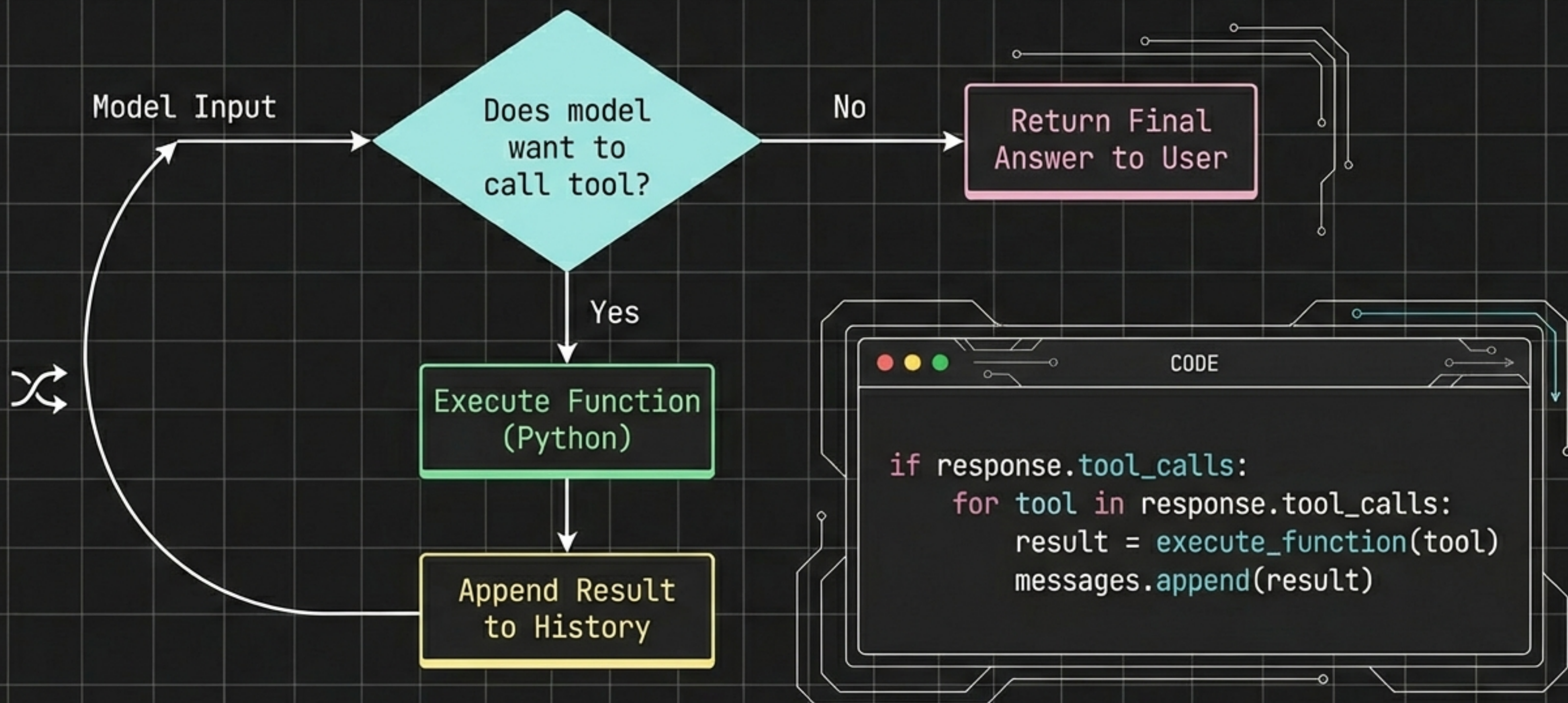
agent = AssistantAgent(tools=[get_weather], model="gpt-4o")
```

The LLM reads this
to understand WHEN
to use the tool.

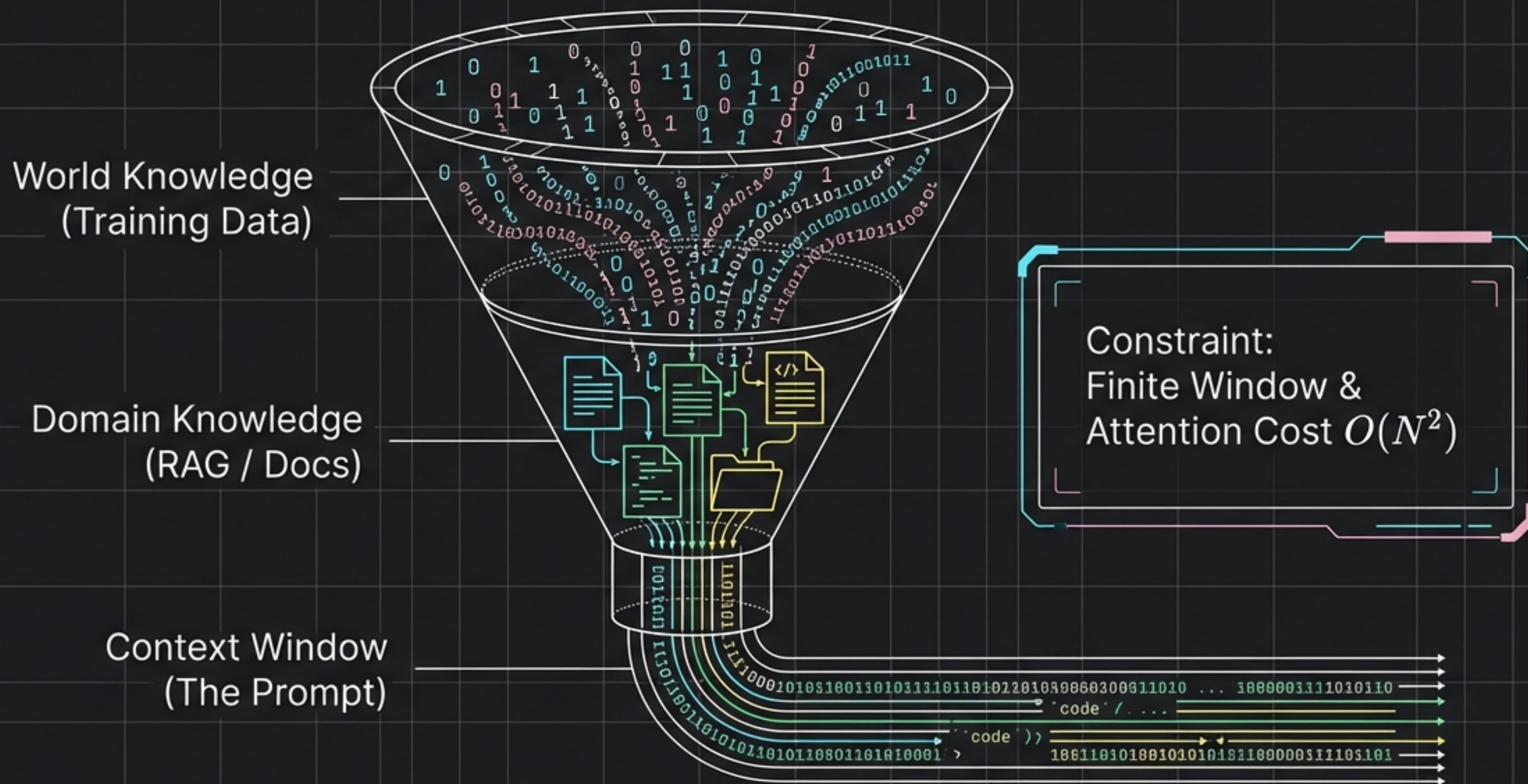
CODE: STATE MANAGEMENT



THE BRAIN: PLANNING LOOP



CONTEXT ENGINEERING SINEERING



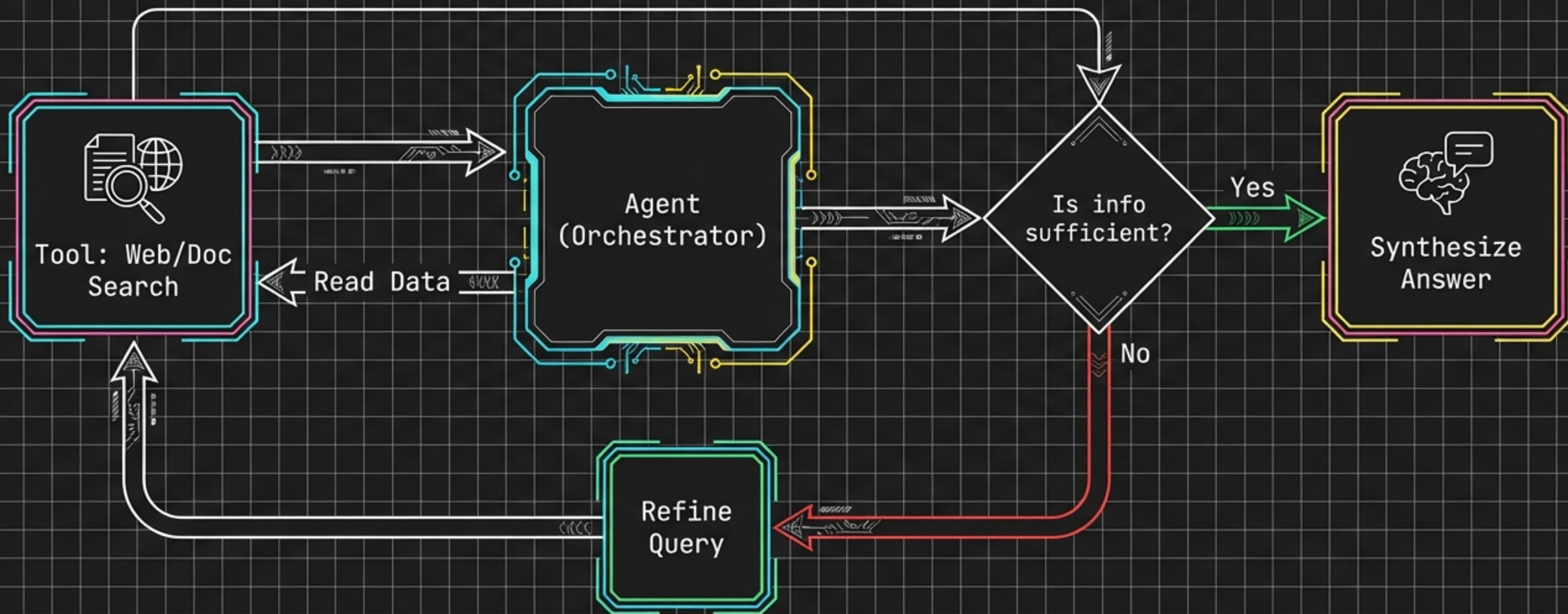
STANDARD RAG

The Open Book Exam Approach (One-Shot)



AGENTIC RAG

The Researcher Approach (Recursive & Self-Correcting)



THE FUTURE: COGNITIVE ARCHITECTURES (CoALA)

Moving from Prompt Engineering to Cognitive System Design.

